COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 009B-CPE21S4	**SECTION:**	CPE21S4
DATE PERFORMED:	4/09/2026	**DATE SUBMITTED:**	4/09/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

main.cpp | QueueLinked.h | StackArray.h

PF > main.cpp > main()

```cpp
#include <iostream>
#include <sstream>
#include <string>
#include "StackArray.h"
#include "QueueLinked.h"

using namespace std;

bool checkSequence(string input) {
    StackArray myStack;
    QueueLinked myQueue;

    int number;
    stringstream ss(input);


    while (ss >> number) {
        myStack.push(number);
        myQueue.enqueue(number);
    }


    while (!myStack.isEmpty() && !myQueue.isEmpty()) {
        int fromStack = myStack.pop();
        int fromQueue = myQueue.dequeue();

        if (fromStack != fromQueue) {
            return false;
        }
    }

    return true;
}

int main() {
    string testCases[] = {
        "101 102 103 102 101",
        "101 102 103",
        "205 205",
        "501",
        "10 20 30 40 50",
        "11 22 33 22 11"
    };

    cout << "Customer Numbers\t\tValid?\n";
    cout << "
```

```
45      cout << "Customer Numbers\t\tValid?\n";
46      cout << "---------------------------\n";
47
48      for (int i = 0; i < 6; i++) {
49          bool result = checkSequence(testCases[i]);
50          cout << testCases[i] << "\t\t";
51
52          if (result == true) {
53              cout << "Yes\n";
54          } else {
55              cout << "No\n";
56          }
57      }
58
59      return 0;
60  }
```

main.cpp | QueueLinked.h | StackArray.h

PF > QueueLinked.h > QueueLinked > dequeue()

```cpp
#ifndef QUEUELINKED_H
#define QUEUELINKED_H

struct Node {
    int data;
    Node* next;
};

class QueueLinked {
private:
    Node* front;
    Node* rear;

public:
    QueueLinked() {
        front = nullptr;
        rear = nullptr;
    }

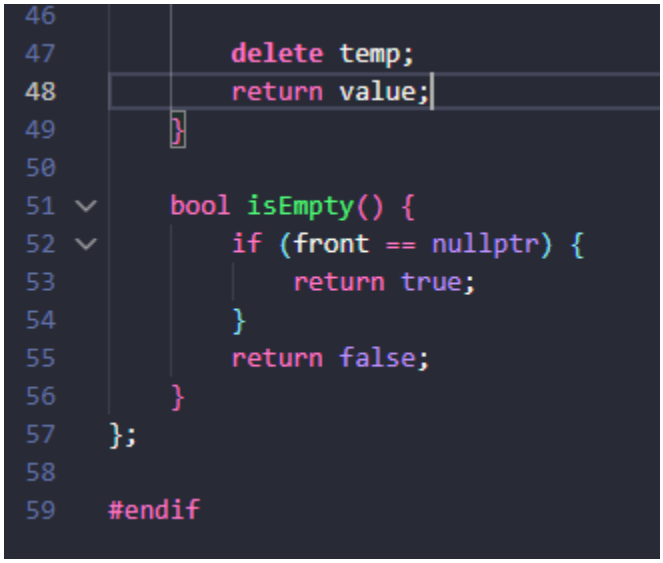
    void enqueue(int num) {
        Node* newNode = new Node;
        newNode->data = num;
        newNode->next = nullptr;

        if (front == nullptr) {
            front = newNode;
            rear = newNode;
        } else {
            rear->next = newNode;
            rear = newNode;
        }
    }

    int dequeue() {
        if (front == nullptr) {
            return -1;
        }

        int value = front->data;
        Node* temp = front;
        front = front->next;

        if (front == nullptr) {
            rear = nullptr;
        }
```



```
46
47          delete temp;
48          return value;
49      }
50
51      bool isEmpty() {
52          if (front == nullptr) {
53              return true;
54          }
55          return false;
56      }
57  };
58
59  #endif
```

main.cpp | QueueLinked.h | StackArray.h

PF > StackArray.h > StackArray > isEmpty()

```h
#ifndef STACKARRAY_H
#define STACKARRAY_H

#define MAX 100

class StackArray {
private:
    int arr[MAX];
    int top;

public:
    StackArray() {
        top = -1;
    }

    void push(int num) {
        if (top < MAX - 1) {
            top++;
            arr[top] = num;
        }
    }

    int pop() {
        if (top >= 0) {
            int value = arr[top];
            top--;
            return value;
        }
        return -1;
    }

    bool isEmpty() {
        if (top < 0) {
            return true;
        }
        return false;
    }
};

#endif
```

```
Problems  Output  Debug Console  Terminal  Ports
Sevilla\OneDrive\Documents\DSA\Prelims\Prelims finals\PF\"main
Customer Numbers          Valid?
---------------------------
101 102 103 102 101       Yes
101 102 103       No
205 205       Yes
501       Yes
10 20 30 40 50        No
11 22 33 22 11        Yes

[Done] exited with code=0 in 0.965 seconds
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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